

GOSFORD HIGH SCHOOL
MATHEMATICS
Extension 2
Assessment task: 30/05/06

Time: 60 minutes.

Question 1.

For the parabola $x^2 + 9y^2 = 9$ find:

- a) The eccentricity. (2)
- b) The length of the major and minor axes. (2)
- c) The co-ordinates of the foci. (1)
- d) The equation of the directrices. (1)
- e) Sketch the ellipse. (2)

Question 2.

- a) By using cylindrical shells find the volume of the solid of revolution when the area enclosed by the graph of $y = e^{2x}$, the 'y' axis and the horizontal line $y = e^2$ is rotated about the 'y' axis. (3)
- b) Calculate the volume when the region between the curve $y = 2x - x^2$, the 'x' axis from $x = 0$, to $x = 2$ is rotated about 'y' axis. (4)

Question 3.

- a) i) Show that the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(a \cos \theta, b \sin \theta)$ is given by $\frac{\cos \theta x}{a} + \frac{\sin \theta y}{b} = 1$ (3)

ii) the tangent at a point $P(a \cos \theta, b \sin \theta)$ on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

cuts the x-axis at M , while the normal at P cuts the x-axis at N .

Prove that $OM \cdot ON = a^2 e^2$, where O is the centre of the ellipse.

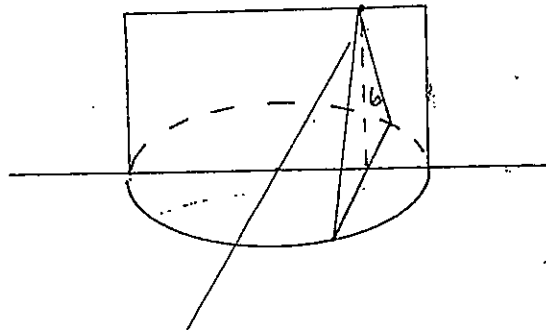
(you may assume the equation of the normal at P is given by

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2).$$
 (3)

- b) Prove that for the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ that the chord of contact from a point on the directrix is a focal chord. (3)

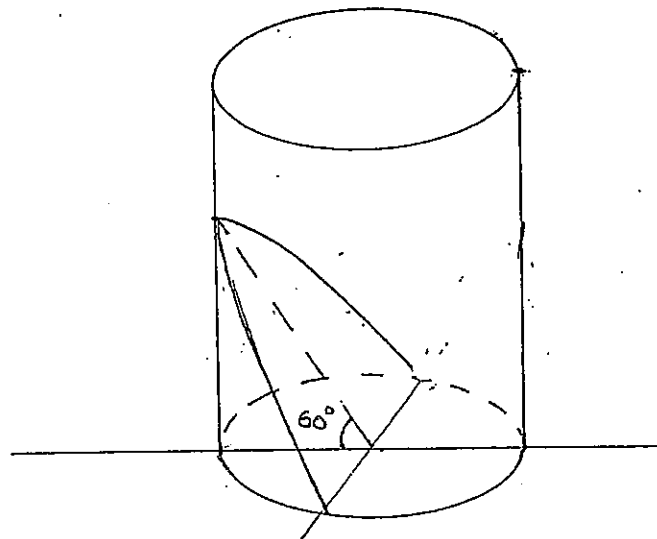
Question 4.

a)



A solid has a base in the form of an ellipse with major axis 10 units and minor axis 8 units. Find the volume if every section perpendicular to the major axis is an isosceles triangle with altitude 6 units. (3)

b)



Two cuts are made on a circular log of radius 8 centimetres, the first perpendicular to the axis of the log and the second inclined at an angle of 60° with the first. If the two cuts meet on a line through the centre, find the volume of the wood cut out. (4)

Q1) $x^2 + 9y^2 = 9$

$$\frac{x^2}{9} + y^2 = 1$$

$a = 3, b = 1$

a) $b^2 = a^2(1 - e^2)$

$$1 = 9(1 - e^2)$$

$$\frac{1}{9} = 1 - e^2$$

$$e^2 = \frac{8}{9}$$

$$e = \frac{2\sqrt{2}}{3}$$

b) major axis = 6
minor axis = 2.

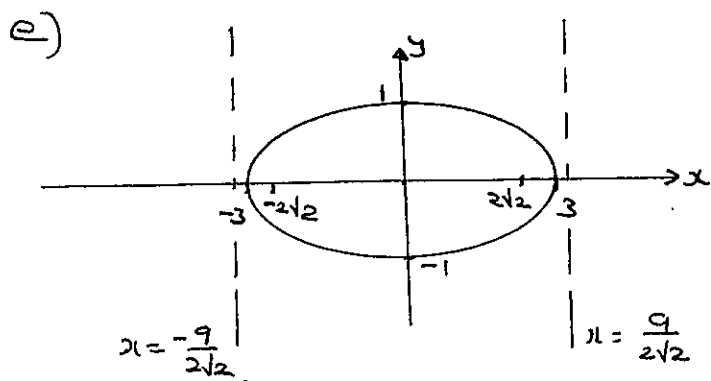
c) foci $(\pm ae, 0)$
 $= (\pm 3 \times \frac{2\sqrt{2}}{3}, 0)$
 $= (\pm 2\sqrt{2}, 0)$

d) directrices.

$$x = \pm \frac{a}{e}$$

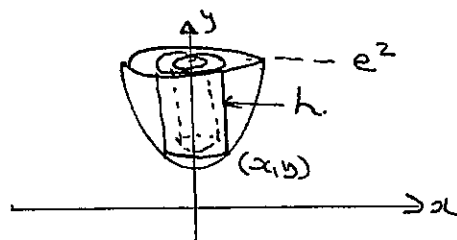
$$= \pm \frac{3}{\frac{2\sqrt{2}}{3}}$$

$$x = \pm \frac{9}{2\sqrt{2}}$$



Q2.)

a)



Volume of shell = $2\pi r h \cdot \Delta x$
 $= 2\pi x (e^2 - y) \Delta x.$

$$\therefore V \doteq \sum_{x=0}^1 2\pi x (e^2 - y) \Delta x$$

$$\therefore V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 2\pi x (e^2 - y) \Delta x$$

$$= 2\pi \int_0^1 x(e^2 - y) dx$$

$$= 2\pi \int_0^1 x(e^2 - e^{2x}) dx$$

$$= 2\pi \int_0^1 x e^2 - x e^{2x} dx.$$

$$= 2\pi \left[\left[\frac{e^2}{2} x^2 \right]_0^1 - \int_0^1 x \cdot \frac{d}{dx} \left(\frac{1}{2} e^{2x} \right) dx \right]$$

$$= 2\pi \left[\frac{e^2}{2} - \left(\frac{x}{2} e^{2x} - \int_0^1 \frac{1}{2} e^{2x} \right) \right]$$

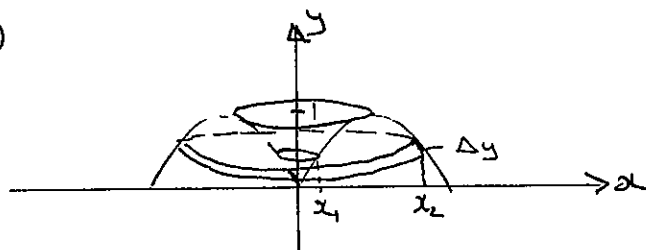
$$= 2\pi \left[\frac{e^2}{2} - \left[\frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} \right]_0^1 \right]$$

$$= 2\pi \left[\frac{e^2}{2} - \left(\frac{e^2}{2} - \frac{e^2}{4} - (0 - \frac{1}{4}) \right) \right]$$

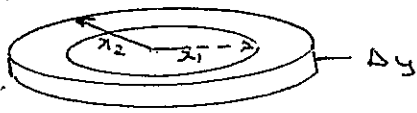
$$= 2\pi \left[\frac{e^2}{4} - \frac{1}{4} \right]$$

$$= \frac{\pi}{2} [e^2 - 1] \text{ cubic units}$$

b)



taking a slice perpendicular to the y axis gives an annulus.



Volume of a Slice = $\pi (x_2^2 - x_1^2) \Delta y$

$$V \doteq \sum_{y=0}^1 \pi (x_2^2 - x_1^2) \Delta y$$

$$V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^1 \pi (x_2^2 - x_1^2) \Delta y$$

$$= \pi \int_0^1 x_2^2 - x_1^2 dy.$$

Now $y = 2x - x^2$

$$x^2 - 2x + y = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4y}}{2}$$

$$= \frac{2 \pm 2\sqrt{1-y}}{2}$$

$$= 1 \pm \sqrt{1-y}$$

$$\therefore x_2 = 1 + \sqrt{1-y}, \quad x_1 = 1 - \sqrt{1-y}$$

$$V = \pi \int_0^1 (1 + \sqrt{1-y})^2 - (1 - \sqrt{1-y})^2 dy$$

$$= \pi \int_0^1 1 + 2\sqrt{1-y} + 1-y - (1 - 2\sqrt{1-y} + 1-y) dy$$

$$= \pi \int_0^1 4\sqrt{1-y} dy$$

$$= 4\pi \int_0^1 \sqrt{1-y} dy$$

$$= 4\pi \left[\frac{-2(1-y)^{3/2}}{3} \right]_0^1$$

$$= 4\pi [0 - (-2/3)]$$

$$= \frac{8\pi}{3} \text{ cubic units}$$

2.

Q3) a) i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

at $(a \cos \theta, b \sin \theta)$

$$\frac{dy}{dx} = \frac{-b^2 a \cos \theta}{a^2 b \sin \theta}$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

$\therefore$ equation of the tangent

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a \sin \theta y - ab \sin^2 \theta = -b \cos \theta x + ab \cos^2 \theta$$

$$b \cos \theta x + a \sin \theta y = ab (\sin^2 \theta + \cos^2 \theta)$$

$$b \cos \theta x + a \sin \theta y = ab$$

$$\frac{\cos \theta x}{a} + \frac{\sin \theta y}{b} = 1.$$

ii) For M put $y = 0$

$$\frac{\cos \theta x}{a} + 0 = 1$$

$$\frac{\cos \theta x}{a} = 1$$

$$x = \frac{a}{\cos \theta}$$

$\therefore M \left(\frac{a}{\cos \theta}, 0 \right)$

eqn of normal at P

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$$

for N, $y = 0$

$$\frac{ax}{\cos \theta} = a^2 - b^2$$

$$x = \frac{(a^2 - b^2) \cos \theta}{a}$$

$$\therefore N \left(\frac{(a^2 - b^2) \cos \theta}{a}, 0 \right)$$

Now OM.ON

$$= \frac{a}{\cos \theta} \times \frac{(a^2 - b^2) \cos \theta}{a}$$

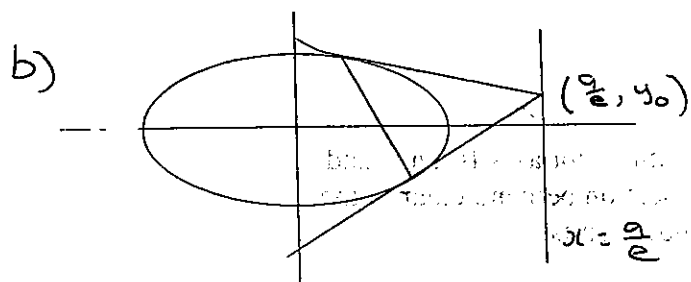
$$= a^2 - b^2$$

$$\text{but } b^2 = a^2(1 - e^2)$$

$$= a^2 - a^2(1 - e^2)$$

$$= a^2 - a^2 + a^2 e^2$$

$$= a^2 e^2$$



Let the point on the directrix be $(\frac{a}{e}, y_0)$

$\therefore$ equation of the chord of contact.

$$\frac{x \cdot \frac{a}{e}}{a^2} + \frac{y y_0}{b^2} = 1$$

$$\frac{xe}{ae} + \frac{yy_0}{b^2} = 1$$

If passes through the focus then $(ae, 0)$ must satisfy the equation of the chord.

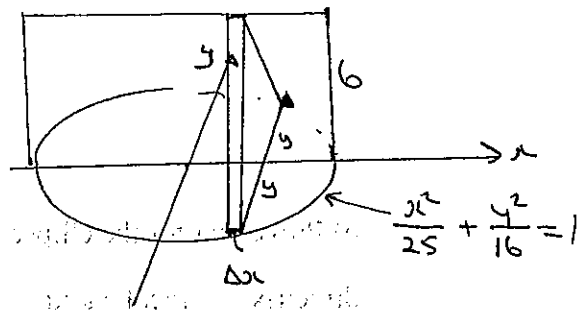
$$\frac{ae}{ae} + 0 = 1$$

$$1 = 1 \quad \text{True}$$

$\therefore$ the chord passes through the focus.

Q4)

a)



$$\text{Volume of a slice} = \frac{1}{2}(2y)6 \cdot \Delta x = 6y \Delta x$$

$$V = \sum_{x=-5}^5 6y \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=-5}^5 6y \Delta x$$

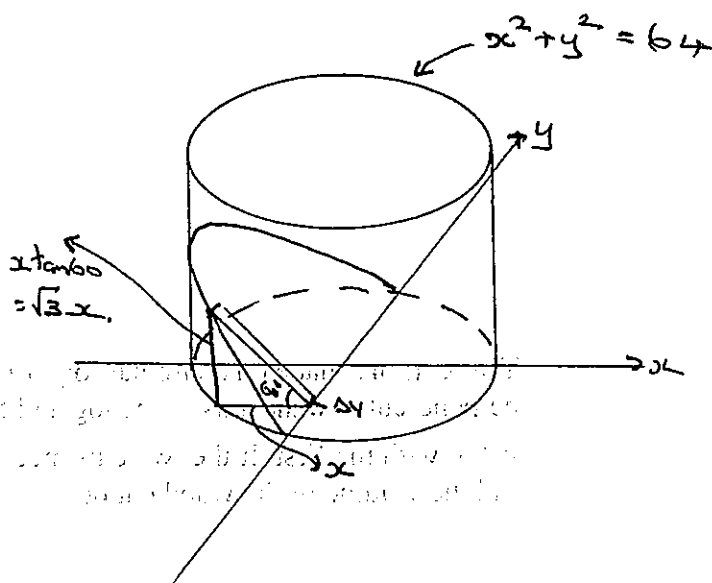
$$= \int_{-5}^5 6y \, dx$$

$$= \int_{-5}^5 6 \left(\frac{4}{5} \sqrt{25 - x^2} \right) dx \quad \left\{ \begin{array}{l} y^2 = 16 \left(1 - \frac{x^2}{25} \right) \\ = 16 \left(\frac{25 - x^2}{25} \right) \\ = \frac{16}{25} (25 - x^2) \\ y = \frac{4}{5} \sqrt{25 - x^2} \end{array} \right.$$

$$= \frac{24}{5} \int_{-5}^5 \sqrt{25 - x^2} \, dx$$

$$= \frac{24}{5} \left(\frac{1}{2} \pi 5^2 \right) \leftarrow (\text{semi-circle})$$

$$= 60\pi \text{ cubic units.}$$



$$\text{Volume of a Slice} = \frac{1}{2} x \times \sqrt{3} x \cdot \Delta y = \frac{\sqrt{3}}{2} x^2 \Delta y$$

(4)

$$V \approx \sum_{y=-8}^8 \frac{\sqrt{3}}{2} x^2 \Delta y$$

$$V \approx \lim_{\Delta y \rightarrow 0} \sum_{y=-8}^8 \frac{\sqrt{3}}{2} x^2 \Delta y$$

$$= \int_{-8}^8 \frac{\sqrt{3}}{2} x^2 dy$$

$$= \sqrt{3} \int_0^8 x^2 dy$$

$$= \sqrt{3} \int_0^8 64 - x^2 dy$$

$$= \sqrt{3} \left[64y - \frac{y^3}{3} \right]_0^8$$

$$= \sqrt{3} \left[\left(512 - \frac{512}{3} \right) - 0 \right]$$

$$= 1024\sqrt{3} \text{ cubic units}$$
